ME2

Voice-controlled smart device

Local wake-word and voice-command recognition with a supervised, evidence-led release boundary

Project scope: supplied WAV or microphone input -> wake gate -> command decoding -> local simulated dashboard action

REPOSITORY-BACKED SUMMARY

What is implemented

16 kHz audio preparation, 40-bin log-Mel features, wake detection, grammar-constrained CTC command decoding, a bounded streaming loop, and a local dashboard.

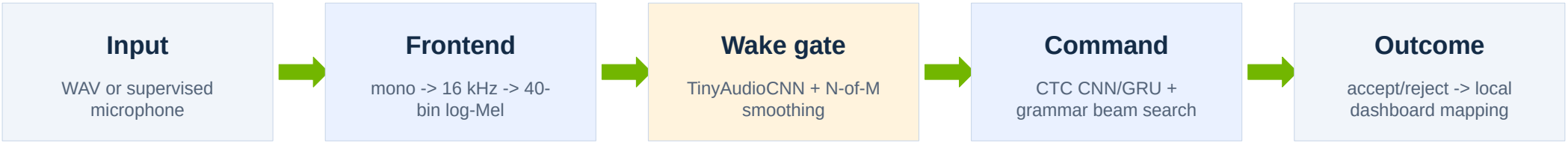
What remains pending

A deployment-ready wake model, fresh comparable wake/OOS evidence, calibrated command rejection, and approval-gated Raspberry Pi 5 validation.

This deck uses repository evidence. It does not claim A100 training, Raspberry Pi 4 validation, ONNX Runtime, causal KV-cache streaming, or production deployment.

Architecture: wake-gated local command recognition

Repository implementation; deck causal/KV-cache terminology is not claimed.



STREAM CONTROLS

Wake windows overlap at the configured hop. A valid wake preserves pre-roll, then command capture observes onset, minimum-speech, endpoint-silence, maximum-duration, queue, and cooldown boundaries.

State	Purpose	Fail-closed behavior
LISTENING_FOR_WAKE	Score overlapping wake windows	No valid wake: no command capture
CAPTURING_COMMAND	Bound command audio after a wake	Silence, timeout, overflow, or overlong audio rejects
DECODING / COOLDOWN	Decode grammar terminal and prevent repeats	Rejected/system result remains non-actionable

Data and model roles

Separate wake and command data contracts protect evaluation boundaries.

Component	Data and preprocessing	Current artifact policy
Wake word	Canonical event: ateko; pronunciation guide: a-teh-ko. Separate manifests retain review, split, and provenance fields.	V5 is supervised primary. V3 is a dashboard fixture and V5-fault-only fallback.
Voice command	HF default splits: 10,733 train, 4,443 test, 202 holdout. Speaker-aware train/validation/calibration separation.	Retained CTC fixture decodes approved grammar terminals; TinyAudioCNN remains a baseline.

MODEL CONSTRAINTS

Wake model

TinyAudioCNN
2,858 parameters
40-bin log-Mel input
Complete ateko only

Command model

Current CTC fixture: CharacterAudioCNN
3,652,572 parameters
Character logits + trie-constrained beam search

Selection boundary

Artifacts have distinct roles.
None is deployment-ready.
Normal inference accepts safe .me2 bundles only.

Evaluation evidence and release boundary

Artifact / split	Observed result	Interpretation
V3 wake **3 positive / 48 negative synthetic test**	0/3 wake events 0/48 false accepts Accuracy 0.9216; macro F1 0.6458	No detected wake event. Retained for dashboard replay and V5-fault diagnostics.
V5 wake **same frozen synthetic test**	2/3 wake events 4/48 false accepts Accuracy 0.9020; macro F1 0.6953	Higher recall than V3, but exceeds the 2/48 false-accept guardrail.
CTC command **4,443-row official test**	Accuracy 0.8911; macro F1 0.8682 20/76 OOS false accepts 316/4,367 in-scope false rejects	Better than baseline, but legacy grammar and uncalibrated acceptance remain blockers.

What these reports do not establish

False activations per hour, robust real-speaker generalization, calibrated command rejection, sustained streaming behavior, or deployment readiness.

Runtime and dashboard workflow

Controls make the operator boundary visible.

CONFIGURATION

Runtime order: built-in defaults -> versioned JSON config -> explicit CLI flag. Artifact audio, vocabulary, grammar, and acceptance contracts remain embedded and validated.

SUPERVISED WORKFLOW

1. Use supplied WAV replay.
2. Check device visibility.
3. Run bounded microphone trials only with an explicit override.
4. Review redacted structured diagnostics.

Mode	What it demonstrates	What it does not prove
CLI WAV replay	Model loading and a decision for supplied files	Microphone, earphone, continuous-stream, or Pi behavior
Dashboard --demo	Synthetic state flow and local simulated actions	Model quality, microphone input, or physical-device control
Live local session	Supervised diagnostic evidence on a chosen device	Unattended operation or deployment approval

Normal runtime logs exclude raw audio. Sensitive debugging can expose private text and paths, so it stays local.

Raspberry Pi 5 validation gates

Current position

The project targets Raspberry Pi 5. The finalization record classifies the work as supervised engineering evaluation, not a production or deployment promotion.

Gate	Required evidence before advancing
Source identity	User-pushed, credential-free remote; supplied branch/tag and exact commit SHA; detached checkout logged on the Pi.
Environment	ARM64/OS/Python/GLIBC confirmation; exact binary wheels, imports, PortAudio, and pip check in an isolated venv.
Artifact and replay	Frozen config and artifact hashes; individual wake and command supplied-WAV checks; integrated replay; non-listening dashboard demo.
Release-quality proof	Separate approved live trials, comparable wake/OOS evidence, calibrated command rejection, sustained measurements, and physical audio validation.

The reference deck's Pi 4 runtime, latency, and real-time-factor cells are blank. This deck does not substitute Pi 5 claims for them.

Submission boundary and evidence references

WHAT CAN BE STATED NOW

ME2 implements local wake-gated command recognition with recorded model artifacts, explicit runtime contracts, a local dashboard, reproducible test commands, and a documented Pi 5 evaluation gate.

WHAT REMAINS TO BE SUPPLIED

Verify publication and license choices, dataset access terms, model release/provenance, training-environment details, and new approved Pi 5 evidence before making a final submission or release claim.

Evidence locations

- README.md: setup, data source, commands, dashboard and stated limits
- docs/MODEL_SELECTION_2026-10-15.md: artifact roles, hashes, selection policy
- docs/V5_FINALIZATION_2026-10-15.md: V5 status and Pi approval gates
- docs/PI5_SSH_SAFE_VALIDATION_2026-10-15.md: historical Pi record and future Git-based setup
- results/*/metrics.json: saved wake, CTC, and baseline evaluation values
- src/me2/: audio, features, bundle loading, streaming, CTC, dashboard implementation

Primary design reference: user-provided ../ppt/me2-deck.pdf, visually reviewed in full. Its placeholders remain unfilled unless repository evidence states otherwise.